

Retrieval-Augmented Generation for Romanian Information Retrieval and Text Mining

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Abstract

The following paper evaluates the performance of a more traditional information retrieval method based on Apache Lucene and the recently introduced family of E5 models (including variations that allow prompting), alongside two ensembles strategies built upon the previously mentioned strategies. The retrieval capabilities are assessed on an original text dataset written in Romanian, a low resources language, alongside a manually annotated set of queries, in terms of success at k, precision at k and recall at k, where k is the number of selected documents.

1 Introduction

As highlighted in a recent survey (Zhu et al., 2024), the rapid growth of focus in advancing information retrieval systems reflects their key role in our day to day lives, through their incorporation in tools such as search engines. The integration of large language models, which continue to drive breakthroughs in the field of natural language processing, into information retrieval systems has become a subject of high interest for researchers. Building on this trend, the current study explores three different approaches: a more traditional information retrieval method, a large language model-based approach, and an ensemble that combines them. An original dataset focused on information about well-known Romanian poets has been curated from public web resources for the evaluation of the proposed methods, alongside a manually annotated set of queries. The paragraphs and queries are written in Romanian, thus the performance will be evaluated on a low-resource language.

2 Related Work

For the retrieval component of Retrieval-Augmented Generation systems, initial approaches (Zhu et al., 2024) relied on term-based

methods and keyword matching. Statistical language models followed, but they were surpassed by neural models which were able to better capture nuances and context. Additionally, the BM25 (Lewis et al., 2021) ranking function achieved impressive results for the estimation of document relevance to a given query. Considering the performance of Large Language Models (LLMs) on complex natural language processing tasks that require reason and generalization capabilities, their potentially beneficial integration in information retrieval systems is a topic of interest being currently explored. The E5 (Embeddings from bidirectional Encoder representations) family of models (Wang et al., 2024) represent state-of-the-art (SOTA) text embeddings that can be used as a general-purpose embedding model on a variety of tasks, including retrieval. One advantage, in comparison to pre-trained models such as GPT, BERT, is that E5 models offer a single vector embedding of texts which can improve efficiency and versatility.

3 Method

In relation to the E5 family of models (Wang et al., 2024), I chose a Zero-shot Retrieval method that implies obtaining the passage embeddings and, for each query, the query embedding. The resulting top k documents are obtained by computing the cosine similarity between query embeddings and passage embeddings. The multilingual-e5 models have been chosen as they provide support for Romanian. For a comparison with a more traditional information retrieval approach which is expected to perform well considering the amount of proper nouns in the relatively small dataset, a method based on Apache Lucene has been implemented.

3.1 Dataset

The dataset is composed out of 83 files that contain paragraphs written in romanian about some of the

most well-known romanian poets and their activity, extracted from various web sources such as news articles and Wikipedia. The files are small in size (ranging from 1KB to 2KB) and the statistics in terms of number of sentences, words and characters are displayed in Table 1. Each file is titled *[author_first_name]_[author_last_name]_[section]_[id]_[optional_sub_id]*, for example, *Mihai_Eminescu_Prezentare_0*; the id range depends on the number of extracted paragraphs from the section of the specific author (for the whole dataset, the upper limit for id can be as low as 0 or as high as 5). Considering the limited amount of data and the high frequency of proper nouns, it can be expected to achieve favorable results utilizing both traditional and large language model-based strategies.

Metric	Min	Max	Average
Sentences	1	16	4.69
Words	27	178	80.33
Characters	173	1055	509.40

Table 1: Dataset text entries statistics

There are 20 unique queries, for which the statistics in terms of number of sentences, words and characters are displayed in Table 2.

Metric	Min	Max	Average
Sentences	1	1	1.0
Words	5	17	9.9
Characters	26	99	57.1

Table 2: Queries text statistics

The queries are rather concise and they have an unique answer, predominantly the name of a well-known romanian poet or the name of one of his works. As expected, considering the theme of the dataset, there are a lot of proper nouns - in both the paragraphs and queries. Although this aspect would usually benefit both approaches, some ambiguity is introduced, for example, for the word "Luceafărul" which, on one hand, is the title of one of Mihai Eminescu's most appreciated poems and, on the other hand, is the name of a popular magazine of that time which a lot of artists have contributed to. Although unique, the answer for a query may be found in multiple documents, depending on the query. For this task, each correctly identified document will be considered equally valuable

as the context it provides is enough to obtain the correct answer - thus, rank awareness is not a concern.

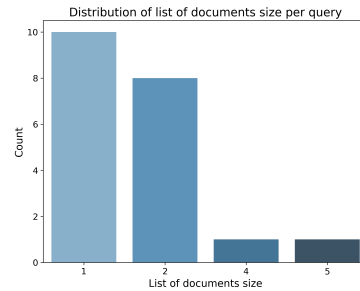


Figure 1: List of documents size per query

3.2 Preprocessing

3.2.1 Apache Lucene-based approach

The following preprocessing methods have been applied: removal of diacritics and stopwords, stemming. As stemming for Romanian is based on diacritics, we have to properly handle keywords with and without diacritics or we risk missing them as their stem may not be found upon the removal of diacritics. A custom solution has been implemented that tokenizes the input using StandardTokenizer, converts it to lowercase, then removes the diacritics using a custom filter, then the stopwords. For each token, augmented alternative forms will be generated based on the possible replacements with diacritics for each character in order to make sure we will be able to identify keywords with and without diacritics. Then, stemming is performed based on RomanianStemmer. The diacritics will be removed once again, followed by the potential duplicates formed (as we have multiple variations with diverse diacritics for the same token, we will need to handle the resulted duplicates after the removal of diacritics). These last steps are added to ensure we are able to handle queries with keywords with or without diacritics. Stopwords are removed after the diacritics, thus the list of stopwords to be removed has been created by removing the diacritics from (ISO, 2024).

3.2.2 E5-based approach

As the E5 (Wang et al., 2024) family of models have been trained on various web resources written in multiple languages - including Romanian - their training data could reflect well the custom Romanian poets dataset which has also been constructed from web resources. I opted to only use the model-associated pretrained tokenizer for preprocessing.

3.3 Method and Results

In terms of metrics, the chosen metric is success at k (S@K) (Armentano et al., 2011) which reflects the probability of finding a correct document among the k selected ones, equal to the percentage of queries for which at least one correct document has been identified. As previously mentioned, we are not concerned in the rank of the documents, thus the lack of rank awareness of the S@K metric does not pose any concerns.

$$S@K = \begin{cases} 1 & \text{if at least one relevant document is in the top } K \\ 0 & \text{otherwise} \end{cases}$$

Precision at k (P@K) and recall at k (R@K) are also displayed for reference, however these metrics may not be able to correctly evaluate the performance due to the nature of the dataset and queries. Precision at k is highly sensitive to the number of relevant documents; for example, some queries only have one relevant document, for a value of 3 for K, the P@K maximum value will be 33.33%.

$$P@K = \frac{\text{Number of relevant documents in } K}{\text{Total number of documents in } K}$$

Similarly, recall at k is sensitive to the number of relevant items.

$$R@K = \frac{\text{Number of relevant documents in } K}{\text{Total number of relevant documents}}$$

The results obtained with the first, more traditional, approach based on Apache Lucene are displayed below.

Model	K	S@K	P@K	R@K
Lucene	1	30.00%	30.00%	20.00%
Lucene	2	50.00%	27.50%	42.50%
Lucene	3	60.00%	21.60%	50.00%
Lucene	4	60.00%	16.25%	50.00%
Lucene	5	60.00%	13.00%	50.33%
Lucene	10	85.00%	10.50%	72.24%
Lucene	20	95.00%	6.33%	80.74%

Table 3: Performance of the Apache Lucene-based approach

Precision is, as expected, decreasing proportionally to the number of selected documents and success and recall are increasing, as the model has a higher chance of identifying relevant documents if asked to retrieve more documents.

Model size	K	S@K	P@K	R@K
small	1	50.00%	50.00%	32.25%
small	2	70.00%	45.00%	54.50%
small	3	90.00%	40.00%	71.75%
small	4	90.00%	32.50%	75.50%
small	5	90.00%	27.00%	78.00%
small	10	96.50%	16.50%	96.50%
small	20	100.00%	8.25%	96.50%

Table 4: Performance of the multilingual-e5-small

Model size	K	S@K	P@K	R@K
base	1	50.00%	50.00%	32.25%
base	2	60.00%	40.00%	44.50%
base	3	85.00%	41.66%	71.75%
base	4	85.00%	33.75%	76.75%
base	5	95.00%	29.00%	86.75%
base	10	95.00%	16.00%	91.50%
base	20	100.00%	8.75%	100.00%

Table 5: Performance of the multilingual-e5-base

The multilingual-e5-small and multilingual-e5-base models perform better than the Apache Lucene based approach which was expected considering that they have been trained on web data - the same source as our original dataset - and the level of ambiguity present in the questions (the previously mentioned example of the word "Luceafărul"). Of note, the larger model does not perform significantly better than the smaller one, with a few exceptions for larger values of K. Due to the comparable performance and reduced size, the multilingual-e5-small is chosen for the following ensemble-based approaches.

For the ensemble methods that combine the initial methods, two strategies are proposed. The first one aims to exploit the performance of Apache Lucene for larger values of K by first selecting Y documents from which it later selects K utilizing the multilingual-e5-small model.

Model	K	S@K	P@K	R@K
Ensemble 1	1	50.00%	50.00%	32.25%
Ensemble 1	2	70.00%	42.50%	55.75%
Ensemble 1	3	85.00%	33.33%	65.75%
Ensemble 1	4	90.00%	27.75%	73.25%
Ensemble 1	5	90.00%	22.00%	73.25%
Ensemble 1	10	95.00%	12.00%	80.74%

Table 6: Performance of Ensemble 1 approach, selecting Y=20 documents using the Apache Lucene method and then K using e5-multilingual-small

This approach led to a clear improvement of the previous results obtained only with Apache Lucene in terms of S@K, P@K, R@K and comparable results with the initial multilingual-e5-small approach.

The second approach involves obtaining the top X documents utilizing both approaches separately and selecting K unique documents based on a computed score r . The score r is computed among the selected documents obtained from both methods by adding the position of the document: for the documents obtained using the e5-multilingual-small model, the first one will have a newly score r of $r = r + 1$, the second $r = r + 2$ and so on; identically for the documents obtained using the other method. The scope is to select the top K documents with the lowest rank; in case of a tie, the documents are arbitraly selected.

Model	K	S@K	P@K	R@K
Ensemble 2	1	25.00%	25.00%	13.75%
Ensemble 2	2	65.00%	32.50%	42.25%
Ensemble 2	3	65.00%	25.00%	45.99%
Ensemble 2	4	70.00%	21.25%	52.00%
Ensemble 2	5	70.00%	16.99%	52.00%
Ensemble 2	10	90.00%	12.00%	75.75%

Table 7: Performance of Ensemble 2 approach, selecting X = 10 documents using the first 2 methods and then K

The performance is slightly better in comparison to the first method, however it does not outperform the previous best results. Next, experiments with smaller values of X have been conducted in an effort to explore the potential improvement of results for smaller values of K.

Model	K	S@K	P@K	R@K
Ensemble 2	1	40.00%	40.00%	22.25%
Ensemble 2	2	70.00%	37.50%	52.25%
Ensemble 2	3	85.00%	36.66%	69.50%
Ensemble 2	4	90.00%	30.00%	77.00%
Ensemble 2	5	95.00%	27.00%	81.75%

Table 8: Performance of Ensemble 2 approach, selecting X = 5 documents using the first 2 methods and then K

This approach surpassed the multilingual-e5-small method for K = 5 (S@5 = 95%, R@5 = 81.75% , P@5 = 27.00%). A similar experiment has been conducted for X = 3, but it failed to outperform the previous results.

Among the conducted experiments, the model e5-multilingual-large-instruct has been utilized alongside a set of prompts such as *"Given a query about well-known Romanian poets, retrieve relevant documents that answer the query"*.

Model	K	S@K	P@K	R@K
large-instruct	1	45.00%	45.00%	28.50%
large-instruct	2	60.00%	30.00%	41.00%
large-instruct	3	65.00%	21.66%	43.49%
large-instruct	4	70.00%	18.75%	47.00%
large-instruct	5	75.00%	16.00%	52.00%
large-instruct	10	75.00%	8.40%	52.00%
large-instruct	20	75.00%	8.20%	52.00%

Table 9: Performance of e5-multilingual-large-instruct with Prompt 1

Model	K	S@K	P@K	R@K
large-instruct	1	45.00%	45.00%	28.50%
large-instruct	2	65.00%	35.00%	44.50%
large-instruct	3	70.00%	24.99%	47.00%
large-instruct	4	70.00%	18.75%	47.00%
large-instruct	5	75.00%	16.00%	52.00%
large-instruct	10	75.00%	8.40%	52.00%
large-instruct	20	75.00%	7.50%	52.00%

Table 10: Performance of e5-multilingual-large-instruct with Prompt 2

The second prompt led to slightly improved results for smaller values of K, however both approaches with e5-multilingual-large-instruct failed to surpass the results of the previously best presented method that relies on a smaller e5 model.

4 Future Work

This initiative could be further developed by creating a larger dataset - potentially a comprehensive database of Romanian authors that could be used by students, teachers and researchers while working on essays and publications - with the use of the associated retrieval models as potential aid. In terms of other approaches that could be developed, experiments utilizing different models (both traditional and LLM-based, potentially more complex), ensembles and diverse prompts could be conducted in order to obtain a more comprehensive understanding of the results. Moreover, LLMs could be integrated such that query-answering could be supported as well - in that sense, the queries set also contains the correct answers.

5 Conclusion

In conclusion, on a limited amount of Romanian text data about well-known national poets collected from the web, the E5 family of models surpasses more traditional approaches, with smaller models leading to the best results. For larger values of K, both of the initial proposed methods perform well, achieving 95% (Apache Lucene), 100% (multilingual-e5-small) values for S@20, out of a total 83 documents. However, the best outcome for a smaller value of K (K = 5) has been achieved by an ensemble method combining the top X documents, separately obtained with the previous methods, and selecting the top K with respect to the lowest computed score r (the score is computed based on the increasing order of appearance in each of the resulting lists, aiming to properly assign priority to the most favorable documents). These findings can be utilized in further research for exploring the performance of the newly introduced E5 models for information retrieval given text data written in low-resource languages or the development of new strategies.

Limitations

A major limitation of the current experiment is the dataset - formed out of only 83 entries - and list of queries - containing only 20 queries - which are written in Romanian, a low resource language, thus the results may not reflect capabilities on other languages. The design, however, is scalable, as the utilized e5-models have support for over 100 languages and other datasets could be easily used

to conduct more experiments. Another limitation to consider is the lack of computational resources - experiments with more complex models (for example, e5_mistral_7b_instruct) could have been conducted if more resources were available.

Ethical Statement

The dataset has been created using publicly available resources from the internet such as news articles which may perpetuate biases. Moreover, considering the nature of the project and the limited amount of data, there may be other well-known authors that have not been mentioned and the sectioned paragraphs may not capture the complete context of the presented information.

References

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A Appendix

List of utilized prompts:

- 1 *Given a query about well-known Romanian poets, retrieve relevant documents that answer the query*
- 2 *Given a romanian poets (web-based information) query, retrieve relevant passages that answer the query*

The passages and queries datasets can be found at:

- <https://www.kaggle.com/datasets/popescuioanalivia/ir-romanian-poets-dataset>
- <https://www.kaggle.com/datasets/popescuioanalivia/ir-queries-romanian-poets-20>

The results obtained with Apache Lucene (K=20) can be found:

- <https://www.kaggle.com/datasets/popescuioanalivia/ir-lucene-romanian-poets-datasets-results-20>